

Security Onion Home Lab — Attack Detection & Incident Response

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Platform: Security Onion 2.4.211 | Oracle VirtualBox | Kali Linux

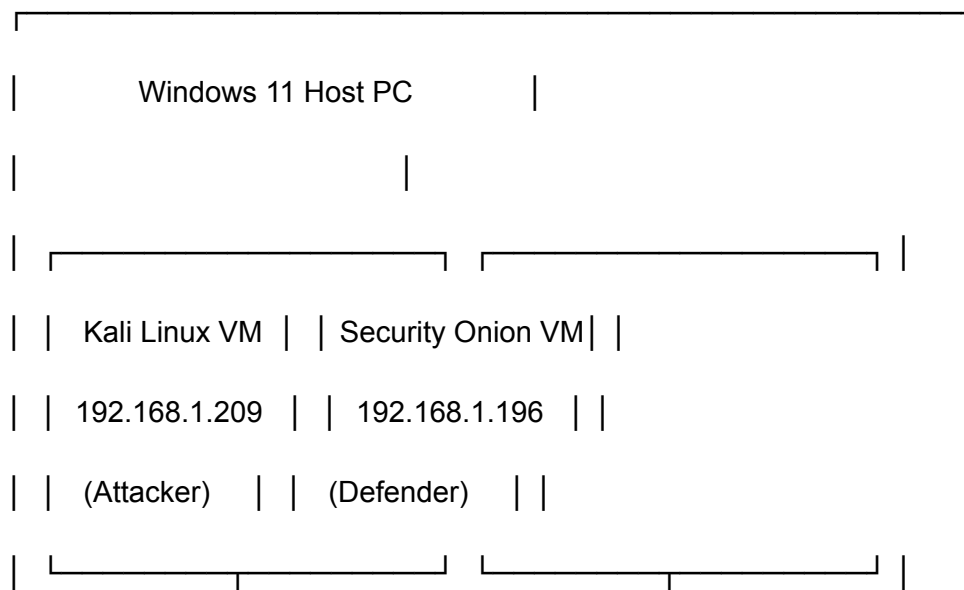
Type: Blue Team | SOC Analysis | Threat Detection

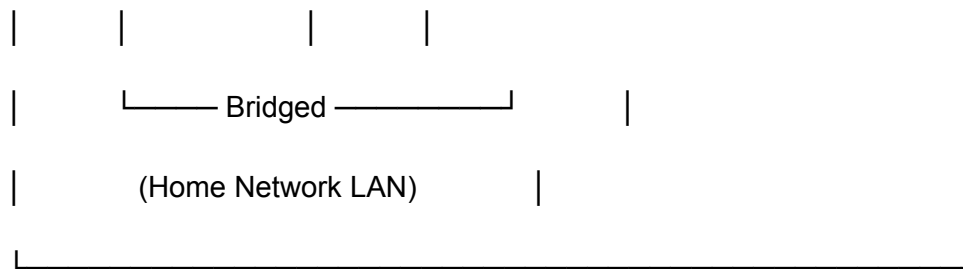
Project Overview

This project demonstrates a full attack-and-detect simulation in a self-built home lab environment. Acting as both attacker (red team) and defender (blue team), I simulated a realistic threat scenario — network reconnaissance followed by a credential brute force attack — and detected, investigated, and documented it using enterprise-grade security tooling.

The result: 404 HIGH severity alerts generated and triaged, Zeek network logs correlated, and a formal SOC incident report produced — end to end.

Lab Architecture





Component	Details
Host OS	Windows 11
Hypervisor	Oracle VirtualBox
Attacker VM	Kali Linux (Bridged Adapter — 192.168.1.209)
Defender VM	Security Onion 2.4.211 EVAL (Bridged Adapter — 192.168.1.196)
Base OS (SO)	Oracle Linux Server 9.7
Detection Engine	Suricata IDS + Sigma Rules + Zeek + Kibana

Project Phases

Phase 1 — Lab Setup & Configuration

- Installed Security Onion 2.4 in EVAL mode on VirtualBox
- Configured bridged networking for host-accessible management IP
- Deployed Kali Linux as attacker VM on same network segment
- Verified bidirectional connectivity between VMs

Phase 2 — Network Reconnaissance (Nmap)

Executed four Nmap scan types from Kali against Security Onion:

Ping sweep — discover live hosts

```
nmap -sn 192.168.1.0/24
```

Aggressive scan — ports, services, scripts, traceroute

```
nmap -A -T4 192.168.1.196
```

```
# Service version detection
```

```
nmap -sV --version-intensity 5 192.168.1.196
```

```
# OS fingerprinting
```

```
nmap -O 192.168.1.196
```

Key findings: Port 22 (SSH/OpenSSH 8.7) open — 999 ports filtered — Linux OS confirmed

Phase 3 — SSH Brute Force (Hydra)

Executed dictionary-based credential attack against SSH using rockyou.txt:

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.196 -t 4 -V -w 10
```

- **~340 attempts** made before manual termination
- **0 valid credentials** found — SSH hardened against attack
- Attack generated significant authentication failure traffic

Phase 4 — Detection & Investigation (Security Onion SOC)

Switched to defender role — investigated all attack activity in Security Onion:

Detection Source	Finding
Sigma Alerts	404 HIGH severity alerts — "Grid Node Login Failure (SSH)"
Zeek DNS Logs	50 DNS query events attributed to attacker IP (192.168.1.209)
Kibana	Cross-source log correlation confirmed full attacker timeline
Alert Timestamps	2026-04-24T22:33:00Z — 2026-04-24T22:37:21Z

Phase 5 — Incident Report

Produced a formal SOC incident report documenting:

- Executive summary and severity assessment
- Full attack timeline with timestamps
- Technical analysis of both attack phases
- Indicators of Compromise (IOC) table
- Detection evidence from all log sources
- Remediation recommendations

 [View Full Incident Report](#)

Indicators of Compromise (IOCs)

IOC Type	Value
Attacker IP	192.168.1.209
Target IP	192.168.1.196
Target Port	22 (TCP/SSH)
Service Targeted	OpenSSH 8.7
Username Attacked	root
Attack Tool — Recon	Nmap 7.95
Attack Tool — Brute Force	Hydra v9.5
Wordlist Used	rockyou.txt
Total Alerts Generated	404 HIGH severity
Attack Window	2026-04-24 21:24 – 22:37 UTC

Tools Used

Offensive (Attacker)

- **Nmap 7.95** — Network reconnaissance and port scanning
- **Hydra v9.5** — SSH credential brute force

- `rockyou.txt` — Dictionary wordlist (14.3M passwords)

Defensive (SOC Analyst)

- `Security Onion 2.4` — SIEM/NSM platform
 - `Suricata` — Network intrusion detection (IDS)
 - `Sigma` — Detection rule engine (404 alerts triggered)
 - `Zeek` — Network traffic analysis and protocol logging
 - `Kibana` — Log visualization and cross-source investigation
 - `Elasticsearch` — Log indexing and search backend
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Troubleshooting Documentation

This project required extensive troubleshooting during the Security Onion setup phase. Every issue encountered is fully documented as a learning reference.

Issue 1 — `VERR_ALREADY_EXISTS` on First Boot

Cause: VirtualBox's Unattended Installation feature auto-generated kickstart files (`ks.cfg` and `.viso`) that conflicted with Security Onion's own installer.

Fix: Deleted `Unattended-*.cfg` and `Unattended-*.viso` files from the VM directory.
Recreated VM with **Skip Unattended Installation** checked.

Prevention: Always check Skip Unattended Installation when creating VMs for Security Onion.

Issue 2 — Forgotten Admin Username

Cause: Admin username forgotten after initial OS install.

Fix: Booted into single-user mode via GRUB editor:

In GRUB, edit the linux line — add to end:

```
rw init=/bin/bash
```

After booting to shell:

```
mount -o remount,rw /
```

```
ls /home          # revealed username
```

```
passwd [username]      # reset password
```

```
sync
```

```
exec /sbin/init        # reboot normally
```

Issue 3 — Installation Failed (Insufficient Disk)

Cause: Virtual disk was 100 GB — below Security Onion's 200 GB minimum.

Fix: Deleted VM entirely and recreated with 200 GB virtual disk.

Lesson: Security Onion stores full PCAPs and ELK data — storage requirements are non-negotiable.

Issue 4 — Web UI Inaccessible (NAT Network Mode)

Cause: Adapter 1 set to NAT assigned an internal `10.0.2.x` address unreachable from the Windows host browser.

Attempted: Port forwarding, Host-Only adapter, `so-allow`, `so-firewall` commands, `iptables` flush — none resolved it.

Fix: Changed Adapter 1 to **Bridged Adapter** (Realtek PCIe GbE Family Controller), then ran:

```
sudo so-ip-update
```

This updated Security Onion's internal configuration to recognize the new bridged IP (`192.168.1.196`).

Issue 5 — Kali VM Could Not Reach Security Onion

Cause: Initial setup used Internal Network adapter type with static IPs, but Security Onion was configured on Bridged adapter — different network segments.

Fix: Changed Kali's adapter to **Bridged Adapter** matching Security Onion's network mode. Both VMs received DHCP addresses from the home router on the same `192.168.1.0/24` subnet.

Incident Report Summary

Incident ID: IR-2026-0424-001

Severity: HIGH

Alerts Generated: 404

Detection Engine: Sigma

Attack Duration: ~73 minutes (21:24 – 22:37 UTC)

Outcome: Attack fully detected — 0 credentials compromised

 [View Full Incident Report](#)

 [View Setup Documentation](#)

Skills Demonstrated

- Network intrusion detection (IDS/IPS)
 - SIEM alert triage and investigation
 - Network traffic analysis (Zeek/Bro)
 - Log correlation across multiple sources (Kibana/Elasticsearch)
 - Threat detection and incident response
 - Network reconnaissance techniques (Nmap)
 - Credential attack simulation (Hydra)
 - Linux system administration (Oracle Linux / Kali)
 - Virtual network architecture design
 - SOC analyst workflow — detection → investigation → documentation
 - Formal incident report writing
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Related Projects

More projects coming soon — check back for updates.

Built and documented by Bryan Sykes | Northern Virginia | 2026

Targeting roles in: SOC Analysis | IT Support | Cybersecurity | Blue Team Operations